PHARMA CARE

A PROJECT REPORT

Submitted by

NALINAA A C

PINGALA PRAGNA A

SRI KARUNYA S S K

THULASI C

in partial fulfilment for the award of the course

AMC1361- MOBILE APPLICATION DEVELOPMENT

IN

DEPARTMENTOF

COMPUTER SCIENCE AND ENGINEERING

(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

K. RAMAKRISHNAN COLLEGE OF ENGINEERING
(AUTONOMOUS)
SAMAYAPURAM, TRICHY

ANNA UNIVERSITY
CHENNAI 600 025

NOVEMBER 2026

PHARMA CARE

A PROJECT REPORT

Submitted by

NALINAA A C (8115U24AM040)
PINGALA PRAGNA A (8115U24AM042)
SRI KARUNYA S S K (8115U24AM052)
THULASI C (8115U24AM061)

in partial fulfilment for the award of the course

AMC1361- MOBILE APPLICATION DEVELOPMENT

DEPARTMENT OF
COMPUTER SCIENCE AND ENGINEERING
(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

Under the Guidance of

Ms.R.SENEGA

Department of CSE (Artificial Intelligence and Machine Learning)
K. RAMAKRISHNAN COLLEGE OF ENGINEERING



K. RAMAKRISHNAN COLLEGE OF ENGINEERING

(AUTONOMOUS)
ANNA UNIVERSITY, CHENNAI



K. RAMAKRISHNAN COLLEGE OF ENGINEERING

(AUTONOMOUS)

ANNA UNIVERSITY, CHENNAI

BONAFIDE CERTIFICATE

Certified that this project report titled **"PHARMA CARE "** is the Bonafide work of **NALINAA A C (8115U24AM040) , PINGALA PRAGNA A (8115U24AM042), SRI KARUNYA S S K (8115U24AM052), THULASI C (8115U24AM061)** who carried out the work under my supervision.

SIGNATURE

Dr. B. KIRAN BALA M.E.,M.B.A.,Ph.D.,

HEAD OF THE DEPARTMENT
ASSOCIATE PROFESSOR,

Department of Computer Science and Engineering (Artificial Intelligence and Machine Learning),
K. Ramakrishnan College of Engineering, (Autonomous)
Samayapuram, Trichy.

SIGNATURE

Ms.R.Senega, M.E.,

SUPERVISOR
ASSISTANT PROFESSOR,

Department of Computer Science and Engineering (Artificial Intelligence and Machine Learning),
K. Ramakrishnan College of Engineering, (Autonomous)
Samayapuram, Trichy.

SIGNATURE OF INTERNAL EXAMINER

NAME:
DATE:

SIGNATURE OF EXTERNAL EXAMINER

NAME:
DATE:

K. RAMAKRISHNAN COLLEGE OF ENGINEERING

(AUTONOMOUS)

ANNA UNIVERSITY, CHENNAI

DECLARATION BY THE CANDIDATES

We declare that, to the best of our knowledge, the work reported here has been composed solely by us and has not been submitted, either in whole or in part, for any previous application for a course.

Submitted for the project Viva-Voce held at K. Ramakrishnan College of Engineering on _____

SIGNATURE OF THE CANDIDATES

Place: Samayapuram
Date:

ACKNOWLEDGEMENT

We thank the almighty GOD, without whom it would not have been possible for us to complete this project.

We wish to address my profound gratitude to **Dr. K. RAMAKRISHNAN**, Chairman, K. Ramakrishnan College of Engineering(Autonomous), who encouraged and gave me all help throughout the course.

We extend my hearty gratitude and thanks to my honorable and grateful Executive Director **Dr.S.KUPPUSAMY, B.Sc., MBA., Ph.D.,** K. Ramakrishnan College of Engineering(Autonomous).

We are glad to thank my Principal **Dr.D.SRINIVASAN, M.E., Ph.D.,FIE., MIIW., MISTE., MISAE., C.Engg,** for giving me permission to carry out this project.

We wish to convey my sincere thanks to **Dr.B.KIRAN BALA, M.E., M.B.A., Ph.D.,** Head of the Department, Computer Science and Engineering(Artificial Intelligence and Machine Learning) for giving me constant encouragement and advice throughout the course.

We are grateful to **Ms.R.SENEGA, B.Tech, M.E., (Ph.D).,Assistant Professor**, Computer Science and Engineering(Artificial Intelligence and Machine Learning), K. Ramakrishnan College of Engineering (Autonomous), for her guidance and valuable suggestions during the course of study.

Finally, We sincerely acknowledged in no less terms all my staff members, my parents and, friends for their co-operation and help at various stages of this project work.

NALINAA A C (8115U24AM040)

PINGALA PRAGNA A (8115U24AM042)

SRI KARUNYA S S K (8115U24AM052)

THULASI C (8115U24AM061)

K. RAMAKRISHNAN COLLEGE OF ENGINEERING
(AUTONOMOUS)
ANNA UNIVERSITY, CHENNAI

DEPARTMENT OF CSE

(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

VISION

To become a renowned hub for AIML technologies to producing highly talented globally recognizable technocrats to meet industrial needs and societal expectation.

MISSION

Mission of the Department

M1 To impart advanced education in AI and Machine Learning, built upon a foundation in Computer Science and Engineering.

M2 To foster Experiential learning equips students with engineering skills to tackle real-world problems.

M3 To promote collaborative innovation in AI, machine learning, and related research and development with industries.

M4 To provide an enjoyable environment for pursuing excellence while upholding strong personal and professional values and ethics.

PROGRAM EDUCATIONAL OBJECTIVES (PEO's)

PEO1 Excel in technical abilities to build intelligent systems in the fields of AI & ML in order to find new opportunities.

PEO2 Embrace new technology to solve real-world problems, whether alone or as a team, while prioritizing ethics and societal benefits.

PEO3 Accept lifelong learning to expand future opportunities in research and product development.

PROGRAM SPECIFIC OUTCOMES (PSO's)

PSO1 Expertise in tailoring ML algorithms and models to excel in designated applications and fields.

PSO2 Ability to conduct research, contributing to machine learning advancements and innovations that tackle emerging societal challenges.

PROGRAM OUTCOMES (POs)

Engineering students will be able to:

PO1 Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems. **PO2 Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3 Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4 Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5 Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

PO6 The Engineer and Society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7 Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO8 Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings

PO9 Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO10 Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO11 Life-long learning: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

ABSTRACT

Pharma Care is a mobile application designed to help users manage their medication schedules efficiently and locate nearby pharmacies with ease. Many individuals, especially elderly people and patients with chronic illnesses, often forget to take their medicines on time or face difficulties in finding nearby pharmacies when needed. It aims to address these challenges by integrating medicine reminders, medication tracking, and location-based pharmacy search into a single application. The application allows users to add medicine details, including the medicine name, dosage, frequency, and duration. It provides automated notifications to remind users to take their medicines at the scheduled time, helping improve medication adherence. In addition, the application uses GPS and map-based services to identify nearby pharmacies and display relevant information such as distance, contact details, operating hours, and navigation directions.Pharma Care incorporates a user-friendly interface, secure database management, and location-based services to provide a convenient and reliable healthcare support system. The application helps users organize their medication routines, reduces the chances of missed doses, and improves access to nearby pharmacy services. Overall, Pharma Care aims to promote better medication management, enhance healthcare accessibility, and support users in maintaining a more organized and responsible approach to their daily health needs.

ABSTRACT WITH POs AND PSOs MAPPING

CO : 5 BUILD A DATABASE FOR REAL – TIME PROBLEMS

ABSTRACT	POs MAPPED	PSOs MAPPED
Pharma Care is a mobile application designed to help users manage their medication schedules effectively and find nearby pharmacies conveniently. Many patients forget to take their medicines on time or face difficulties in managing multiple prescriptions, which may affect their health. It aims to address these problems through automated medicine reminders, medication tracking, and location-based pharmacy search features. The application allows users to add medicine details, including medicine name, dosage, frequency, and duration. It generates timely notifications to remind users to take their medicines according to their schedules. Additionally, the application uses GPS and map-based services to identify nearby pharmacies and display their distance, contact details, working hours, and navigation directions.	**PO1 – 2** **PO2 – 3** **PO3 – 3** **PO4 – 2** **PO5 – 3** **PO6 – 2** **PO7 – 2** **PO8 – 2** **PO9 – 2** **PO11 –2**	**PSO1 – 1** **PSO2 – 1**

Note: 1- Low, 2-Medium, 3- High

TABLE OF CONTENTS

CHAPTER No.		TITLE	PAGE No.
		ABSTRACT	**viii**
		LIST OF ABBREVIATIONS	**xii**
		LIST OF FIGURES	**xiii**
		LIST OF TABLES	**xiv**
1		**INTRODUCTION**	**1**
	1.1	Introduction	1
	1.2	Objectives	2
	1.3	Purpose and Importance	3
	1.4	Data Source Description	3
	1.5	Project Summarization	4
2		**LITERATURE SURVEY**	**5**
	2.1	Traditional Medicine Purchase and Management	5
	2.2	Existing Mobile Health Solutions	5
	2.3	Location-Based Services (LBS)	6
	2.4	Review of Related Research Papers	6
	2.5	Summary of Literature Survey	10
3		**SYSTEM DESIGN**	**11**
	3.1	Proposed Workflow	11
	3.2	Architecture Diagram	11
	3.3	Technology Stack	13
	3.4	Database Design	14
	3.5	Application Pages	14
4		**RELEVANCE OF THE PROJECT**	**16**
	4.1	Addressing User Needs	16
	4.2	Enhancement to Medicine Management	16
	4.3	Benefits and Limitations	17

5		**MODULE DESCRIPTION**	**18**
	5.1	User Module	18
	5.2	Medicine Search Module	20
	5.3	Prescription Upload Module	21
	5.4	Medicine Order Module	23
	5.5	Medicine Reminder Module	24
	5.6	Nearby Pharmacy Module	26
	5.7	Admin Module	28
6		**RESULTS AND DISCUSSION**	**30**
	6.1	Result	30
	6.2	Discussion	31
7		**TESTING, VALIDATION AND QUALITY ASSURANCE**	**32**
	7.1	Introduction to Testing	32
	7.2	Testing Strategy	32
	7.3	Test Cases	33
	7.4	Validation	35
	7.5	Quality Assurance	36
	7.6	Defect Tracking and Resolution	36
	7.7	Summary of Testing	37
8		**CONCLUSION AND FUTURE SCOPE**	**38**
	8.1	Conclusion	38
	8.2	Future Scope	39
		APPENDICES	**40**
		Appendix A - Source Code	40
		Appendix B - Screenshots	42
		REFERENCES	**45**

LIST OF ABBREVIATIONS

S.NO	ACRONYM	ABBREVIATION
1	API	Application Programming Interface
2	DB	Database
3	EHR	Electronic Health Records
4	FCM	Firebase Cloud Messaging
5	GPS	Global Positioning System
6	LBS	Location-Based Services
7	OTP	One Time Password
8	SMS	Short Message Service
9	UI	User Interface
10	UX	User Experience

LIST OF FIGURES

FIGURE NO.	TITLE	PAGE NO.
3.2.1	Architecture Diagram of PharmaCare	12
5.1.1	Flow Diagram of User Module	19
5.2.1	Flow Diagram of Medicine Search Module	21
5.3.1	Flow Diagram of Prescription Upload Module	22
5.4.1	Flow Diagram of Medicine Order Module	24
5.5.1	Flow Diagram of Medicine Reminder Module	26
5.6.1	Flow Diagram of Nearby Pharmacy Module	27
5.7.1	Flow Diagram of Admin Module	29
B.1	Login Page	42
B.2	Registration Page	42
B.3	Registration Page (Scrolled View)	42
B.4	Login Page with Credentials Entered	42
B.5	Home Page	43
B.6	Search Medicines Page	43
B.7	My Cart Page	43
B.8	Medicine Reminders Page	43
B.9	User Profile Page	44
B.10	Prescription History Page	44
B.11	Upload Prescription Page	44
B.12	Selecting a Prescription Image from the Gallery	44

LIST OF TABLES

TABLE NO.	TITLE	PAGE NO.
2.5.1	Summary Of The Reviewed Literature	10
3.3.1	Technologies Used In Pharmacare	13
3.4.1	Database Tables Of Pharmacare	14
7.2.1	Levels Of Testing	33
7.3.1	Test Cases Of Pharmacare	35

CHAPTER 1
INTRODUCTION

1.1 INTRODUCTION

Healthcare services are rapidly moving towards digital platforms, and the mobile phone has become the most convenient medium through which people access them. Buying medicines, however, still depends largely on visiting a local medical store, carrying a physical prescription and remembering the exact time at which every medicine has to be taken. For elderly people, patients with chronic diseases and working professionals with limited time, this routine is inconvenient and often leads to missed doses, delayed purchases and irregular treatment.

"PharmaCare – Smart Medicine Care Mobile Application" is a cross-platform healthcare application developed using Flutter and Firebase that combines online pharmacy services with personal medicine management. The application allows users to search for medicines, upload a doctor's prescription, order medicines online, track their orders and set daily medication reminders from the comfort of their homes. It also helps users find nearby pharmacies using their current location and provides navigation support through maps.

The system has two types of users: the customer, who uses the mobile application to search, order and manage medicines, and the administrator (pharmacist), who manages the medicine catalogue, verifies prescriptions, processes orders and monitors reports. By bringing these services into a single application, PharmaCare provides a convenient, secure and user-friendly digital healthcare companion.

1.2 OBJECTIVES

The primary objective of PharmaCare is to provide a convenient, secure and user-friendly platform for managing medicines and healthcare needs. The application helps users purchase medicines online, track their orders and never miss a daily medication through reminder notifications. The specific objectives of the project are listed below.

- To develop a Flutter-based mobile application that allows users to register, log in and manage their profile and delivery addresses securely.
- To enable users to search for medicines by name, browse them by category and view details such as price, dosage and availability.
- To allow users to upload a doctor's prescription using the camera or gallery and to have it verified by the pharmacist.
- To provide a complete ordering system with a shopping cart, multiple payment options (UPI, Cards and Cash on Delivery) and real-time order tracking.
- To implement a medicine reminder system that sends timely notifications and records missed doses and medication history.
- To integrate location-based services that help users find nearby pharmacies, view their contact details and navigate to them.
- To provide an administrator module for managing medicines, categories, orders, prescriptions, user accounts and reports.

Overall, the application aims to deliver a user-friendly, accessible and efficient healthcare solution that enhances medication adherence, reduces the effort involved in purchasing medicines and improves the quality of life of patients, caregivers and anyone who relies on timely medication and quick access to pharmaceutical services.

1.3 PURPOSE AND IMPORTANCE

The purpose of PharmaCare is to provide a dependable digital healthcare companion that assists users in obtaining and managing their medicines with accuracy and ease. Many individuals, especially elderly patients, chronic illness sufferers and busy professionals, find it difficult to visit a pharmacy regularly, to keep their prescriptions safe and to remember the correct time for each dose. This can lead to poor treatment outcomes and avoidable health complications. PharmaCare addresses these challenges by offering online ordering, digital prescription storage and a smart reminder system in a single application.

The application is also created to give users quick access to essential pharmaceutical services whenever the need arises. By using the device location, the system helps users identify nearby pharmacies, view their contact details and navigate to them, which is especially important in urgent situations, during late-night requirements or when the user is in an unfamiliar area. The combination of online pharmacy services, medication reminders and location support makes the application a complete solution for medicine purchase and medicine management.

1.4 DATA SOURCE DESCRIPTION

The data used in PharmaCare originates from a combination of user-generated information, administrator-managed information and external service providers. The information entered by users forms the foundation of the application, and the accuracy of reminders, orders and prescription records depends on the completeness of this data. The main data sources are described below.

- **User data:** name, e-mail address, mobile number, password (handled by Firebase Authentication) and delivery addresses entered at the time of registration and profile management.
- **Medicine catalogue:** medicine names, categories (tablets, syrups, capsules, etc.), prices, dosage details and availability entered and updated by the administrator.
- **Prescription data:** images of doctors' prescriptions captured through the camera or selected from the gallery and their verification status.
- **Order and payment data:** cart items, order status, payment method and transaction status generated while ordering medicines.
- **Reminder data:** medicine name, dosage, reminder times, days and the taken or missed status of every dose entered by the user.
- **Location data:** the current GPS location of the device and pharmacy details (name, address, distance, contact number) obtained from the Google Maps Platform.

1.5 PROJECT SUMMARIZATION

PharmaCare is a comprehensive mobile-based healthcare solution designed to simplify the way users purchase and manage their medicines. The project integrates user authentication, medicine search, prescription upload, online ordering, medicine reminders and a nearby pharmacy locator into a single platform, supported by an administrator module for managing the entire system. The user interface is developed using Flutter and Dart, while Firebase provides authentication, database, storage and notification services.

The system enables users to find medicines quickly, upload prescriptions securely, place and track orders, and receive timely reminders for each dose. At the same time, it enables the pharmacist to verify prescriptions, process orders and analyse reports.

CHAPTER 2
LITERATURE SURVEY

2.1 TRADITIONAL MEDICINE PURCHASE AND MANAGEMENT

Traditional medicine purchase and management practices depend largely on manual methods. Patients visit a nearby medical store with a paper prescription, wait in a queue, and buy the medicines prescribed by the doctor. Handwritten notes, calendars, pillboxes and verbal instructions from doctors are then used to remember when each medicine has to be taken, and family members often remind elderly or chronically ill persons to take their medicines on time.

Although these methods have served their purpose for many years, they have significant limitations in today's fast-paced lifestyle. Paper prescriptions can be lost or become unreadable, medicines may not be available in the nearest store, and late-night or emergency requirements are difficult to fulfil. Patients with chronic illnesses such as diabetes, hypertension or heart disease often receive multiple prescriptions, which makes a complex regimen hard to follow from memory. Manual systems also provide no way of monitoring adherence, so neither the patient nor the healthcare provider can track missed doses accurately. These limitations highlight the need for a digital alternative that offers online purchase, secure prescription storage, reminders and real-time tracking.

2.2 EXISTING MOBILE HEALTH SOLUTIONS

With the growth of smartphones and digital healthcare, many mobile health (mHealth) applications have been developed to help users manage their medicines. Most of them fall into two groups. The first group consists of medication reminder applications that offer alarms, dose tracking and simple medicine lists. The second group consists of online pharmacy applications that allow users to search for medicines and order them for home delivery.

In practice, these two groups of applications work separately. Reminder applications rarely allow the user to purchase medicines or upload a prescription, while ordering applications do not help the user to take the medicines on time. Very few applications combine prescription verification, order tracking, reminders, missed-dose records and a nearby pharmacy finder in one place. As a result, users have to install and manage several applications for different parts of their healthcare routine, which reduces convenience and increases the chance of errors.

2.3 LOCATION-BASED SERVICES (LBS)

Location-Based Services provide location-aware information in real time and are widely used in healthcare applications that help users identify nearby hospitals, clinics, emergency facilities, blood banks and pharmacies. Modern smartphones are equipped with GPS and map services that make it possible to detect the user's position, search for places within a given radius and guide the user to the destination with step-by-step navigation.

In a pharmacy application, location services are most useful when the user needs a medicine urgently, when the medicine is out of stock in the online catalogue, or when the user is travelling. By integrating the Google Maps Platform, PharmaCare uses these services to list nearby pharmacies with their distance, contact details and map navigation.

2.4 REVIEW OF RELATED RESEARCH PAPERS

To understand the importance of medication adherence and the role of mobile applications in healthcare, six research papers and reports were studied. A brief review of each is given below.

2.4.1 Adherence to Medication

Osterberg and Blaschke (2005) reviewed the problem of medication non-adherence in the New England Journal of Medicine. The authors explain that when patients do not take their medicines as prescribed, the benefits of treatment are lost and the result is avoidable complications, repeated hospital visits and higher healthcare costs. The common reasons discussed include forgetfulness, complex dosage schedules, poor understanding of the treatment and difficulty in obtaining medicines. The paper concludes that simplifying regimens and supporting patients with reminders and easy access to medicines improves adherence.

Relevance to PharmaCare: The findings support the need for a reminder module and for convenient online access to medicines within the same application.

2.4.2 Adherence to Long-Term Therapies: Evidence for Action

The World Health Organization report edited by Sabaté (2003) studies adherence to long-term therapies in chronic diseases. It reports that adherence among patients with chronic illness is only about half in developed countries and is generally lower in developing countries. The report describes adherence as a multi-dimensional problem influenced by social and economic factors, the health system, the condition, the therapy and the patient, and stresses that improving adherence can have a greater effect on public health than the development of new treatments.

Relevance to PharmaCare: The report justifies a patient-centred application that reduces access barriers and provides continuous support through reminders and medication history.

2.4.3 Smartphone Medication Adherence Apps: Potential Benefits to Patients and Providers

Dayer et al. (2013) examined smartphone applications that are designed to improve medication adherence. The study concludes that such applications have the potential to benefit both patients and healthcare providers through features such as dose reminders, tracking of doses and access to medicine information. However, the authors also observed that the available applications differ widely in the features that they provide and that many of them offer only basic functions.

Relevance to PharmaCare: The limited functionality of existing applications motivates PharmaCare to combine reminders with prescription upload, ordering and a pharmacy locator.

2.4.4 Mobile Phone Apps to Improve Medication Adherence: A Systematic Stratified Review

Santo et al. (2016) performed a systematic stratified review of studies that evaluated mobile phone applications for improving medication adherence. Many of the reviewed studies reported improvements in adherence when applications with reminders and tracking were used. At the same time, the authors point out that the evidence is limited by small studies and short follow-up periods and recommend that applications should be designed carefully and tested properly before being used by patients.

Relevance to PharmaCare: The review supports the design of the reminder module and highlights the importance of the testing, validation and quality assurance activities described in Chapter 7.

2.4.5 Beyond Self-Tracking and Reminders: Designing Smartphone Apps That Support Habit Formation

Stawarz, Cox and Blandford (2015) analysed medication reminder applications and argued that reminders and self-tracking alone are not enough to help people build lasting medication habits. They suggest that applications should fit into the user's daily routine and allow flexibility, for example by letting the user respond to a reminder, postpone it or record a missed dose, so that the application remains useful over a long period.

Relevance to PharmaCare: The reminder module of PharmaCare therefore provides taken, snooze and missed-dose options together with a medication history, instead of a simple alarm.

2.4.6 How Smartphones Are Changing the Face of Mobile and Participatory Healthcare

Kamel Boulos et al. (2011) present an overview of how smartphones are changing mobile and participatory healthcare, with the eCAALYX project as an example. The paper explains that built-in features of smartphones such as the camera and GPS allow applications to provide context-aware and location-aware health services. It also draws attention to issues such as privacy, data security and regulation that must be considered when health information is handled by mobile applications.

Relevance to PharmaCare: The paper supports the use of the camera for prescription capture and GPS for the nearby pharmacy module, and it emphasises the secure handling of health data.

2.5 SUMMARY OF LITERATURE SURVEY

The studied papers clearly show that poor medication adherence is a serious healthcare problem and that mobile applications can help to improve it. A summary of the papers and their relevance to PharmaCare is given in Table 2.5.1

S.NO.	Authors and Year	Focus Area	Relevance to PharmaCare
[1]	Osterberg and Blaschke, 2005	Causes and consequences of medication non-adherence	Need for reminders and easy access to medicines
[2]	Sabaté (Ed.), WHO, 2003	Adherence to long-term therapies	Patient-centred support for chronic patients
[3]	Dayer et al., 2013	Smartphone adherence applications	Combining reminders with other services
[4]	Santo et al., 2016	Systematic review of adherence apps	Reminder design and need for proper testing
[5]	Stawarz, Cox and Blandford, 2015	Habit formation and reminders	Taken, snooze and missed-dose options
[6]	Kamel Boulos et al., 2011	Smartphones in mobile healthcare	Camera, GPS and secure data handling

Table 2.5.1 Summary of the Reviewed Literature

The review also shows a clear research gap. Existing applications and studies concentrate mainly on reminders, while online medicine purchase, prescription verification, order tracking and pharmacy location are provided by separate applications. PharmaCare fills this gap by integrating all of these services into a single, easy-to-use mobile application supported by an administrator module.

CHAPTER 3
SYSTEM DESIGN

3.1 PROPOSED SYSTEM

The proposed workflow of PharmaCare begins with user registration and secure authentication, which allows every individual to create a personal profile and add delivery addresses. After logging in, the user reaches the Home Page, from where the medicine search, prescription upload, shopping cart, order tracking, reminder and nearby pharmacy features can be accessed.

To purchase medicines, the user searches for a medicine by name or category, views its details and adds it to the cart. If the medicine requires a doctor's prescription, the user uploads the prescription through the camera or gallery and the pharmacist verifies it. The user then selects the delivery address and a payment method (UPI, Card or Cash on Delivery), places the order and tracks its status in real time until delivery.

Independently of ordering, the user can create daily medicine reminders. The reminder engine sends a notification at the scheduled time, records whether the dose was taken or missed and maintains the medication history. When medicines are needed urgently, the nearby pharmacy locator uses the device GPS and map services to show pharmacies around the user with their contact details and navigation. All data, including user profiles, medicines, orders, prescriptions, payments and reminders, is stored securely in the cloud database so that it can be retrieved and synchronised across sessions.

3.2 ARCHITECTURE DIAGRAM

The overall architecture of PharmaCare is shown in Figure 3.2.1. The customer interacts with the Flutter mobile application, which contains the six user-side modules. The administrator uses the Admin Module to manage the system. The application communicates with Firebase for authentication, database, file storage and push notifications, with the Google Maps Platform for location and pharmacy data, and with a payment gateway for online payments.

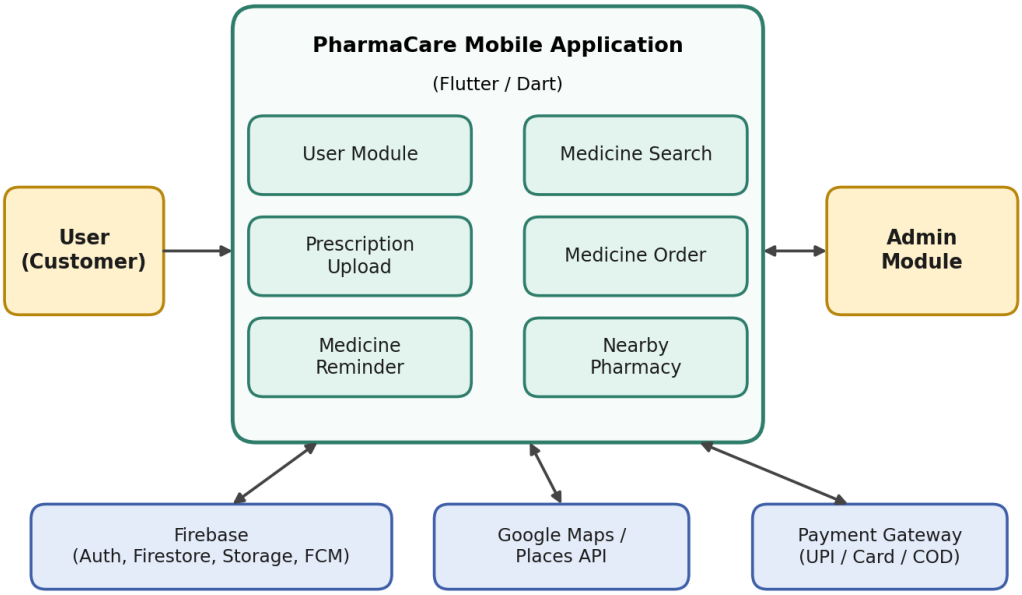


Figure 3.2.1 Architecture Diagram of PharmaCare

The architecture follows a layered approach. The presentation layer consists of the Flutter screens and widgets. The business logic layer handles validation, cart calculation, reminder scheduling and role checks. The cloud layer provides Firebase Authentication, Cloud Firestore, Cloud Storage and Firebase Cloud Messaging. External services such as the Google Maps Platform and the payment gateway are accessed through their respective SDKs and APIs. This separation makes the application scalable and easy to maintain.

3.3 TECHNOLOGY STACK

The technologies used for developing PharmaCare are summarised in Table 3.3.1.

Layer	Technology	Purpose
Frontend	Flutter, Dart	Cross-platform user interface and application logic
Backend	Firebase, Node.js (optional)	Authentication, notifications and optional custom server logic
Database	Firebase Cloud Firestore (MySQL as alternative)	Storage of users, medicines, orders, prescriptions, payments and reminders
File Storage	Firebase Cloud Storage	Secure storage of prescription images
Location and Maps	Device GPS, Google Maps Platform	Nearby pharmacy search and navigation
Notifications	Firebase Cloud Messaging, local notifications	Medicine reminders and order updates
Development Tool	Android Studio	Coding, debugging and testing on emulators and devices

Table 3.3.1 Technologies Used in PharmaCare

3.4 DATABASE DESIGN

The application stores its data in seven main tables (collections), which are described in Table 3.4.1.

Table	Important Fields	Description
Users	user_id, name, email, phone, address, role	Profile and address details of customers and administrators
Medicines	medicine_id, name, category_id, price, dosage, stock, prescription_required	Medicine catalogue managed by the administrator
Categories	category_id, category_name	Medicine categories such as tablets, syrups and capsules
Orders	order_id, user_id, items, total_amount, address, status, order_date	Customer orders and their current status
Prescriptions	prescription_id, user_id, image_url, status, uploaded_at	Uploaded prescriptions and verification status
Payments	payment_id, order_id, method, amount, status	Payment details of every order
Reminders	reminder_id, user_id, medicine_name, time, days, status	Reminder schedule and dose history

Table 3.4.1 Database Tables of PharmaCare

3.5 APPLICATION PAGES

The user interface of PharmaCare consists of the following pages, which are designed using Flutter widgets and Material Design guidelines.

- **Splash Screen:** displays the application logo and checks whether the user is already logged in.
- **Login and Sign Up:** allows a new user to register and an existing user to sign in.
- **Home Page:** provides quick access to search, prescriptions, orders, reminders and nearby pharmacies.
- **Search Medicine and Medicine Details:** allow searching, browsing and viewing complete medicine information.
- **Shopping Cart and Order Page:** allow the user to manage the cart, pay and track orders.
- **Prescription Upload:** allows the user to upload and view prescriptions.
- **Reminder Page:** allows the user to create reminders and view the medication history.
- **Profile Page:** allows the user to manage personal information and delivery addresses.

CHAPTER 4
RELEVANCE OF THE PROJECT

4.1 ADDRESSING USER REQUIREMENTS

PharmaCare effectively addresses the common challenges faced by users in obtaining and managing their medicines. Elderly people and patients with chronic illnesses often find it difficult to visit a pharmacy regularly, while working professionals have little time to stand in queues. The application allows them to search for medicines and order them from home, with delivery to their doorstep and a choice of payment methods.

Many users also forget to take their medicines on time or lose their paper prescriptions. PharmaCare solves these problems by providing clear and timely reminders and by storing uploaded prescriptions securely in the cloud. In urgent situations, where a medicine is required immediately, the nearby pharmacy module helps the user to find and reach the closest pharmacy without delay.

4.2 ENHANCEMENT TO MEDICINE MANAGEMENT

The application significantly enhances medicine management by bringing purchase, prescription handling and daily consumption into one platform. Users can keep their prescriptions, orders and reminders together, track the status of every order and view the history of the doses they have taken or missed. This organised record helps users to follow their treatment plans consistently and supports them in managing their health independently.

For the pharmacist, the Admin Module replaces manual registers with a digital catalogue, a prescription verification queue and an order management screen. Reports and analytics help the administrator to understand sales and stock levels, which improves the efficiency and reliability of the pharmacy service.

4.3 BENEFITS AND LIMITATIONS

PharmaCare offers several key benefits: a user-friendly interface that can be used by people of all age groups, a single application for ordering medicines and managing reminders, secure cloud storage of prescriptions and orders, real-time order tracking, and location-based access to nearby pharmacies. By automating reminders and prescription handling, the application reduces the risk of missed doses and the effort of buying medicines.

The system also has certain limitations. Searching medicines, placing orders and finding nearby pharmacies require an active internet connection, and the accuracy of GPS-based results may vary depending on the device and network conditions. The effectiveness of reminders depends on notification permissions being enabled and on battery-saving settings of the device not blocking them. Prescription verification depends on the availability of the pharmacist, and the delivery time depends on the pharmacy’s delivery arrangements.

CHAPTER 5
MODULE DESCRIPTION

PharmaCare consists of seven modules: the User Module, Medicine Search Module, Prescription Upload Module, Medicine Order Module, Medicine Reminder Module, Nearby Pharmacy Module and Admin Module. Each module is explained in detail below along with a diagram that shows its working.

5.1 USER MODULE

The User Module is the entry point of PharmaCare and is responsible for identifying every person who uses the system. It manages registration, login, profile maintenance, delivery address management and personal information, and ensures that sensitive data such as prescriptions, orders and reminders is linked only to the authorised account holder. The module is implemented using the Splash, Login and Sign Up, and Profile pages of the application, which are connected to Firebase Authentication and the Users collection of Cloud Firestore.

When the application is opened, the Splash Screen checks whether a valid session already exists. If it does, the user is taken directly to the Home Page; otherwise the Login and Sign Up page is displayed. During registration the user enters a name, e-mail address, mobile number and password. These inputs are validated on the device (valid e-mail format, minimum password length and mandatory fields) and are then sent to Firebase Authentication, which creates the account and returns a unique user ID. A matching document is created in the Users collection to store the profile details. The main functions of the module are listed below.

- **Registration and Login:** creation of a new account and secure sign-in using e-mail and password.

- **Profile Management:** viewing and editing of the name, mobile number and other profile details.
- **Address Management:** adding, updating and deleting delivery addresses and selecting the default address.
- **Personal Information:** maintenance of personal details that are required for delivery and communication.
- **Session Handling and Password Recovery:** persistent login for convenience and password reset through e-mail.

Security is a key concern in this module. Passwords are never stored in the application database because they are handled entirely by Firebase Authentication, and Firestore Security Rules ensure that a user can read and modify only his or her own document. The Admin Module uses the same authentication service, but an additional role field is checked so that administrator screens cannot be opened by normal users.

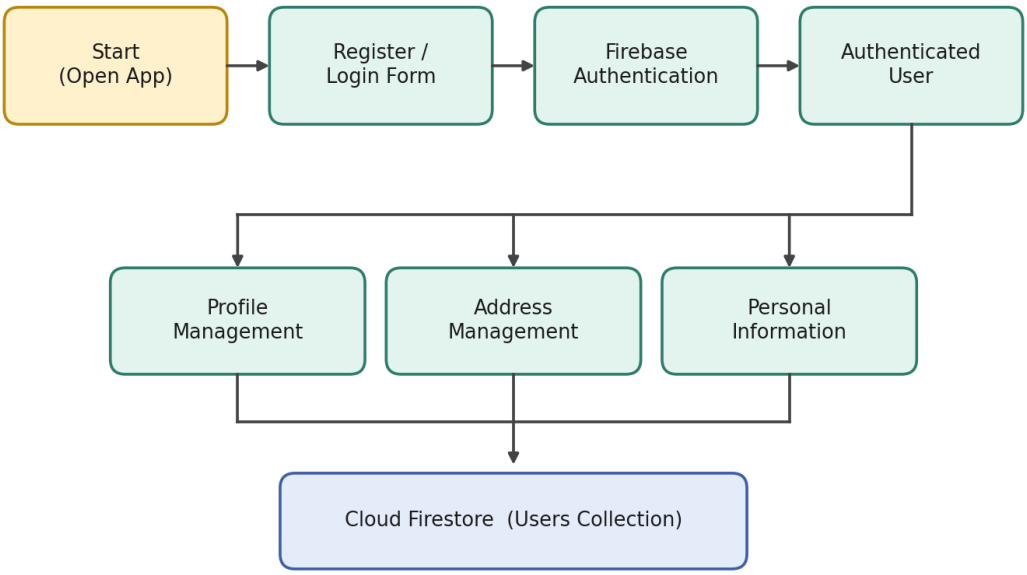


Figure 5.1.1 Flow Diagram of User

As shown in Figure 5.1.1, the user opens the application and submits the registration or login form. Firebase Authentication verifies the credentials and, once the user is authenticated, access is granted to the profile, address and personal information sections. All changes made in these sections are saved in the Users collection of Cloud Firestore.

5.2 MEDICINE SEARCH MODULE

The Medicine Search Module helps users to find the medicines they need quickly. It allows the user to search for a medicine by name, to browse medicines by category such as Tablets, Syrups and Capsules, and to view complete details of a selected medicine. The module reads its data from the Medicines and Categories collections of Cloud Firestore, which are maintained by the administrator through the Admin Module.

When the user types in the search box, the application sends a query to Firestore that matches the entered text against a lower-case search field stored with every medicine, so that the search is not affected by capital letters. The matching medicines are displayed as a list with their name, price and availability. When a category is selected, the application filters the Medicines collection using the category ID and shows only the medicines of that category. The main functions of the module are given below.

- **Search by Name:** instant search with suggestions while the user types.
- **Browse by Category:** category chips on the Home Page and Search page for filtering medicines.
- **Medicine Details:** display of name, category, price, dosage, description and availability.
- **Availability Check:** an In Stock or Out of Stock label based on the stock value in the database.

- **Prescription Indicator:** a label that tells the user if the medicine can be supplied only against a prescription.
- **Add to Cart:** direct transfer of the selected medicine to the Medicine Order Module.

Because the catalogue is stored in the cloud, any change made by the administrator, such as a new price or a change in stock, is reflected in the application without the user having to update it.

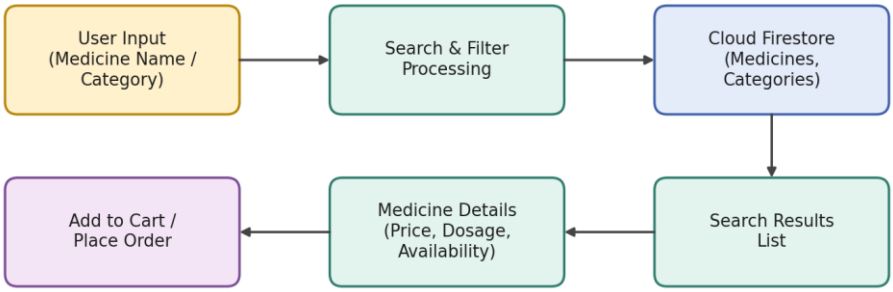


Figure 5.2.1 Flow Diagram of Medicine Search

Figure 5.2.1 shows that the user input is processed by the search and filter logic, which queries Cloud Firestore. The returned results are displayed as a list, and when the user selects a medicine its details are shown and the user can add it to the cart or proceed to order.

5.3 PRESCRIPTION UPLOAD MODULE

The Prescription Upload Module allows users to submit the doctor's prescription digitally instead of carrying a paper copy to the pharmacy. The user can capture the prescription using the camera or select an existing image from the gallery. The uploaded prescription is verified by the pharmacist, and verified records are stored securely so that they can be used again for future orders.

After the image is selected, the application validates the file type and size and compresses the image to reduce the upload time. The image is then uploaded to Firebase Cloud Storage in a folder that belongs to the user, and a record containing the image link, the user ID, the upload time and the status Pending is created in the Prescriptions collection. The pharmacist opens the Prescription verification screen of the Admin Module, checks the doctor's details, the date and the prescribed medicines, and marks the prescription as Verified or Rejected with a remark. The user is informed of the decision through a push notification sent by Firebase Cloud Messaging. The main functions of the module are listed below.

- **Upload through Camera or Gallery:** flexible selection of the prescription image.
- **File Validation:** checking of the file type and size before the upload.
- **Pharmacist Verification:** review of every prescription and update of its status.
- **Status Notification:** notification of the user about approval or rejection.
- **Secure Storage and History:** storage of prescription records that are accessible only to the owner and the administrator.

  Storage security rules ensure that a prescription image can be read only by the user who uploaded it and by the administrator, which protects the privacy of the patient.

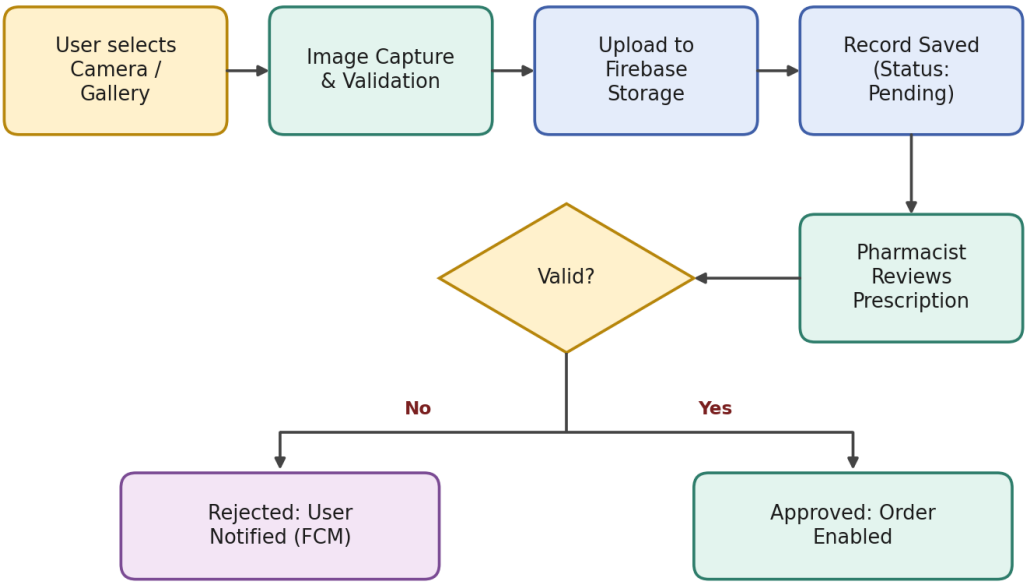


Figure 5.3.1 Flow Diagram of Prescription Upload

As illustrated in Figure 5.3.1, the prescription is captured, validated and uploaded to Firebase Storage, and a pending record is saved. The pharmacist reviews the prescription and, if it is valid, the order of the prescribed medicines is enabled; otherwise the prescription is rejected and the user is notified so that a correct prescription can be uploaded.

5.4 MEDICINE ORDER MODULE

The Medicine Order Module provides the online purchase facility of PharmaCare. It consists of the shopping cart, checkout, payment and order tracking functions. The module connects the user, the pharmacist and the payment service, and it keeps the user informed about the progress of every order.

The user adds medicines to the cart from the Medicine Search Module and can change the quantity or remove an item. The cart total is calculated automatically. At checkout the user selects a delivery address and one of the available payment options: UPI, Cards or Cash on Delivery. If the cart contains a medicine that requires a prescription, the application checks that a verified prescription is available before the order is accepted. After a successful payment, or after confirmation of Cash on Delivery, an order document is created in the Orders collection and a payment record is created in the Payments collection. The stock of the ordered medicines is reduced using a Firestore transaction so that the quantity remains correct even if several users order at the same time.

- **Shopping Cart:** adding, removing and changing the quantity of medicines with an automatic total.
- **Checkout:** selection of the delivery address and review of the order summary.
- **Multiple Payment Options:** UPI, Cards and Cash on Delivery.
- **Real-Time Order Tracking:** order status changes from Placed to Packed, Out for Delivery and Delivered, and the user sees each change immediately.
- **Order History:** a record of all previous orders of the user.

The administrator updates the status of every order from the Admin Module. Since the application listens to the Orders collection, the user sees the updated status on the Order Page in real time.

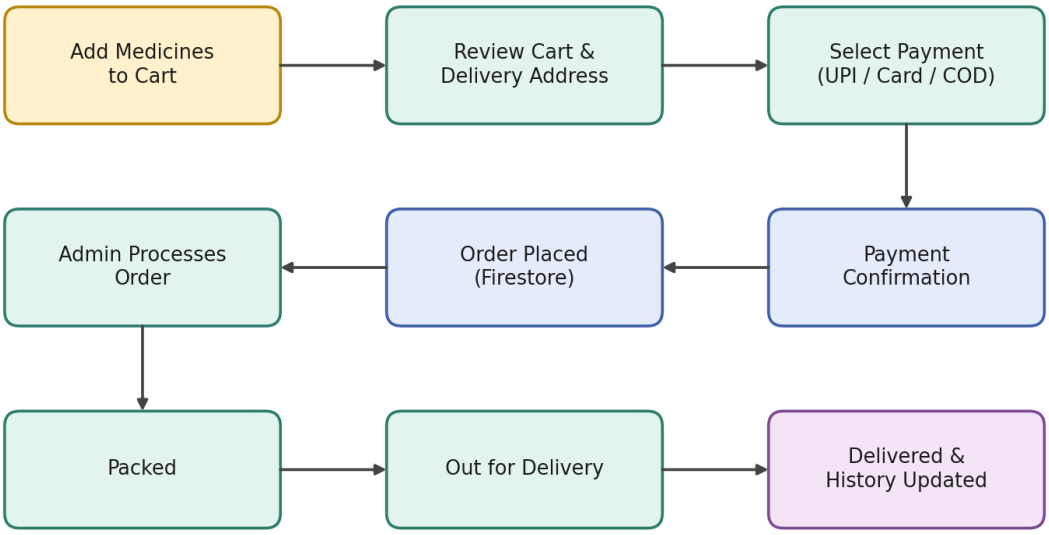


Figure 5.4.1 Flow Diagram of Medicine Order

Figure 5.4.1 shows the order flow. After the cart is reviewed and the payment method is selected, the payment is confirmed and the order is stored in Firestore. The administrator processes the order, which then moves through the packed, out for delivery and delivered stages, after which the order history of the user is updated.

5.5 MEDICINE REMINDER MODULE

The Medicine Reminder Module helps users to take their medicines at the right time. It allows the user to set daily reminders, receive notification alerts and track missed doses and the medication history. This module is important for elderly patients and for people who have to take several medicines every day.

To create a reminder, the user enters the medicine name, dosage, one or more reminder times, the days of the week, the start and end dates and optional notes. The reminder is saved in the Reminders collection of Firestore and is also scheduled on the device using local notifications, so that the alert is shown at the correct time even when the application is closed or the device is offline. When the alert appears, the user can mark the dose as Taken or postpone it using the snooze option. If the user does not respond within the allowed time, the dose

is recorded as Missed and a follow-up notification is sent through Firebase Cloud Messaging. The reminders are rescheduled automatically when the device is restarted.

- **Daily Reminders:** reminders with multiple times per day and selected days of the week.
- **Notification Alerts:** sound, on-screen notification and optional vibration for every dose.
- **Taken and Snooze Options:** quick actions that can be performed directly from the notification.
- **Missed Dose Tracking:** automatic recording of doses that were not marked as taken, with a follow-up alert.
- **Medication History:** a record of taken and missed doses that helps the user to review adherence.

On recent versions of Android, the module requests the notification permission and the permission to schedule exact alarms, so that the reminders are delivered on time.

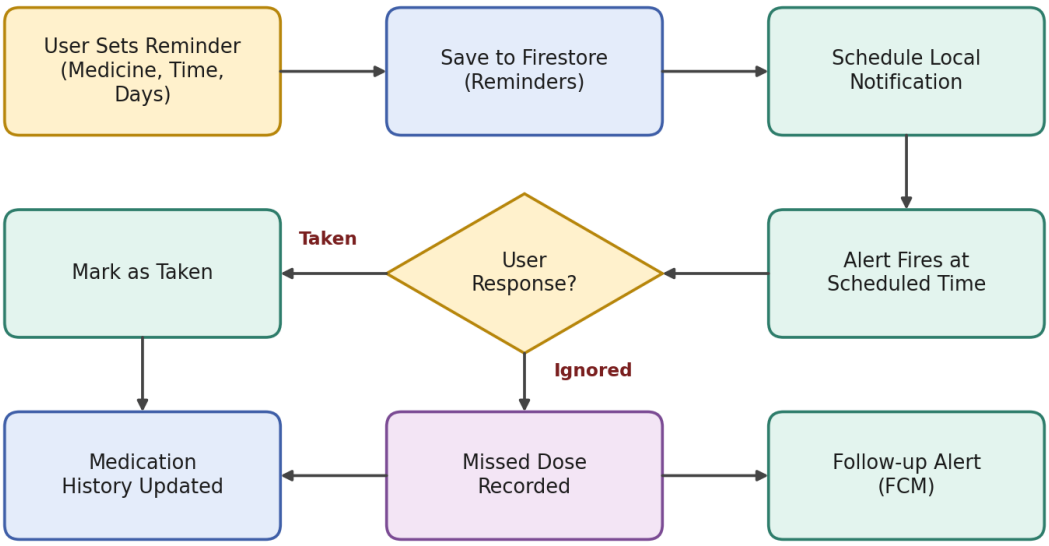


Figure 5.5.1 Flow Diagram of Medicine Reminder

Figure 5.5.1 explains the working of the module. The reminder is saved and scheduled, the alert fires at the scheduled time, and the response of the user decides the next step: a taken dose updates the medication history, while an ignored alert is recorded as a missed dose and triggers a follow-up notification.

5.6 NEARBY PHARMACY MODULE

The Nearby Pharmacy Module helps the user to locate pharmacies around the current location. It is particularly useful in emergencies, during late-night requirements or when a medicine is not available in the online catalogue. The module uses the device GPS, the Google Maps Platform and the Places API to display the pharmacies and to guide the user to the selected one.

When the user opens the module, the application asks for the location permission. After the permission is granted, the current coordinates of the device are obtained and sent to the Google Places API with the type set to pharmacy and a search radius. The returned results contain the name, address, distance, rating, opening status and contact number of every pharmacy. The pharmacies are displayed as markers on an interactive map and as a list. When the user selects a pharmacy, its details are displayed and the user can call the pharmacy or start navigation in the maps application.

- **Location Detection:** accurate detection of the current position using GPS.
- **Pharmacy Search:** retrieval of nearby pharmacies using the Places API.
- **Map View:** display of pharmacies as markers on an interactive map.
- **Pharmacy Details and Contact:** name, address, distance and contact number with a call option.
- **Navigation Support:** step-by-step directions to the selected pharmacy.

If the user denies the location permission, the module displays a message that asks the user to enable location access and allows the user to search for a place manually.

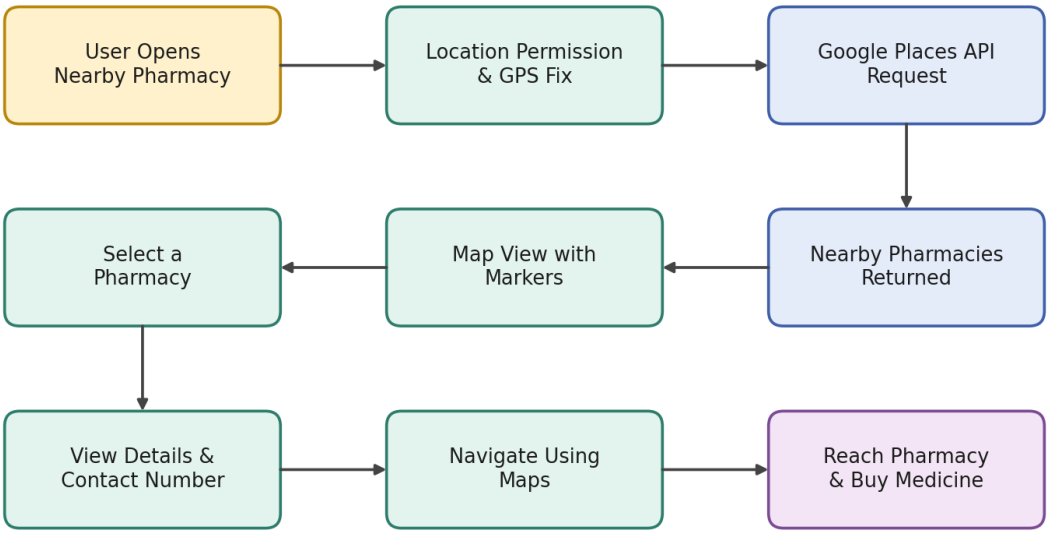


Figure 5.6.1 Flow Diagram of Nearby Pharmacy

As seen in Figure 5.6.1, the GPS location is sent to the Google Places API, the returned pharmacies are displayed on the map, and the selected pharmacy can be contacted or reached through the navigation feature.

5.7 ADMIN MODULE

The Admin Module is used by the administrator or pharmacist to manage the complete system. It is accessible only to authorised administrators, whose accounts are identified by a role field that is checked after login. The module reduces the manual work of the pharmacy and keeps the data used by the customer application accurate and up to date.

Through the Admin Dashboard, the administrator can add, update and delete medicines and manage the medicine categories, which immediately updates the catalogue that users see. The administrator verifies the prescriptions uploaded by users, processes customer orders by changing their status, manages user accounts and generates reports. All these operations are performed on the Firestore collections and on Firebase Storage. The main functions of the module are listed below.

- **Manage Medicines:** adding, updating and deleting medicines with price, dosage and stock.
- **Manage Categories:** creating and editing categories such as tablets, syrups and capsules.
- **Process Orders:** viewing new orders and updating their status up to delivery.
- **Verify Prescriptions:** approving or rejecting uploaded prescriptions with remarks.
- **Manage User Accounts:** viewing and controlling the accounts of registered users.
- **Reports and Analytics:** summaries of orders, sales, low-stock medicines and pending prescriptions.

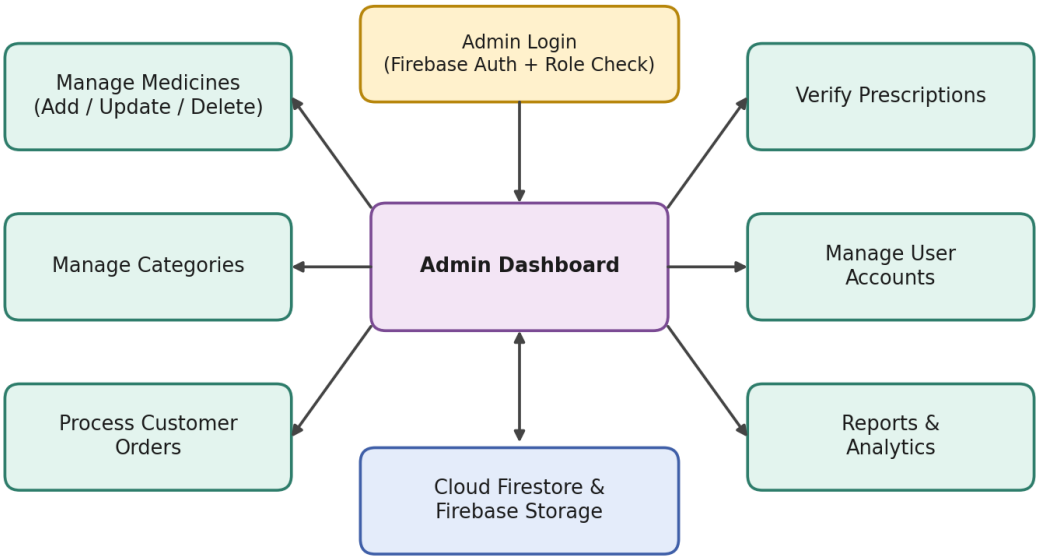


Figure 5.7.1 Flow Diagram of Admin

Figure 5.7.1 shows that the administrator first logs in and is checked for the admin role. The Admin Dashboard then provides access to medicine management, category management, order processing, prescription verification, user management and reports, all of which read from and write to Cloud Firestore and Firebase Storage.

CHAPTER 6
RESULTS AND DISCUSSION

6.1 RESULT

PharmaCare was developed using Flutter and Firebase and was tested to evaluate its performance, usability and effectiveness in supporting medicine purchase and daily medication management. The application performed well in all its core functions. The module-wise results are summarised below.

- **User Module:** users were able to register, log in and manage their profile and delivery addresses without difficulty.
- **Medicine Search Module:** medicines could be found by name and by category, and the details, price and availability were displayed correctly.
- **Prescription Upload Module:** prescriptions were uploaded through the camera and gallery and the verification status was updated after the pharmacist's decision.
- **Medicine Order Module:** items could be added to the cart, orders could be placed with the available payment options and the order status was updated in real time.
- **Medicine Reminder Module:** reminder notifications were triggered at the scheduled times and taken and missed doses were recorded in the history.
- **Nearby Pharmacy Module:** nearby pharmacies were displayed on the map with their details and navigation support.
- **Admin Module:** the administrator was able to manage medicines, categories, orders, prescriptions and users.

6.2 DISCUSSION

The results show that PharmaCare successfully combines online pharmacy services with medication management in a single application. Users can find and order medicines without visiting a store, keep their prescriptions in digital form and receive reminders that help them to follow their schedules more consistently. The nearby pharmacy feature gives additional support in urgent situations, and the Admin Module makes it easy for the pharmacist to keep the catalogue and orders under control.

The system depends on internet connectivity and on the accuracy of GPS and notification settings of the device, but it still offers a practical and convenient solution for everyday healthcare needs. Overall, the application demonstrates how an integrated platform can improve medication adherence and the accessibility of pharmacy services.

CHAPTER 7
TESTING, VALIDATION AND QUALITY ASSURANCE

7.1 INTRODUCTION TO TESTING

Since PharmaCare deals with medicines, prescriptions and payments, it is essential that the application works correctly, protects user data and gives accurate information. Testing, validation and quality assurance were therefore carried out throughout the development of the project. Testing checks whether the application behaves as expected, validation checks whether the data entered by the user and the results produced by the application are correct, and quality assurance ensures that the application meets standards of performance, security and usability.

7.2 TESTING STRATEGY

A multi-level testing strategy was followed, starting from small units of code and ending with testing by real users. The levels of testing, their scope and the tools used are given in Table 7.2.1.

Testing Level	Scope	Tools Used
Unit Testing	Individual functions such as input validators, cart total calculation and data models	Flutter test package
Widget Testing	Screens, forms, buttons and navigation of the application	Flutter widget tester
Integration Testing	Interaction of the application with Firebase, Google Maps and notifications	integration_test, Firebase Local Emulator Suite
System Testing	Complete flows such as search, order and track	Android emulator and physical devices

Testing Level	Scope	Tools Used
Compatibility Testing	Different Android versions and screen sizes	Android Studio virtual devices
User Acceptance Testing	Usability and correctness as judged by sample users and the pharmacist	Feedback from users

Table 7.2.1 Levels of Testing

7.3 TEST CASES

Selected test cases that were prepared for the modules of PharmaCare, together with the expected results, are listed in Table 7.3.1.

Test ID	Module	Test Scenario	Expected Result
TC01	User	Register with a valid name, e-mail and password	Account is created and the Home Page is opened
TC02	User	Register with an e-mail that is already used	Error message is displayed
TC03	User	Log in with a wrong password	Login is rejected with an error message
TC04	User	Update profile and delivery address	Changes are saved and shown on the Profile Page
TC05	Search	Search for an available medicine	Matching medicines are displayed with price and availability
TC06	Search	Search for a medicine that is not in the catalogue	A “no medicines found” message is displayed

Test ID	Module	Test Scenario	Expected Result
TC07	Search	Select a category such as Syrups	Only medicines of that category are listed
TC08	Prescription	Upload a valid prescription image from the gallery	Image is uploaded and the status is shown as Pending
TC09	Prescription	Upload an unsupported or very large file	Upload is blocked with a validation message
TC10	Prescription	Pharmacist rejects a prescription	Status becomes Rejected and the user is notified
TC11	Order	Add an item to the cart and change its quantity	Cart total is recalculated correctly
TC12	Order	Place an order using Cash on Delivery	Order is created with the status Placed
TC13	Order	Cancel or fail an online payment	Order is not confirmed and the user can retry
TC14	Order	Administrator changes the order status	Updated status is shown to the user in real time
TC15	Reminder	Create a daily reminder for a given time	Notification is shown at the scheduled time
TC16	Reminder	Ignore a reminder notification	Dose is recorded as missed and a follow-up alert is sent
TC17	Pharmacy	Allow the location permission	Nearby pharmacies are shown on the map
TC18	Pharmacy	Deny the location permission	A message asks the user to enable location; manual search is available

Test ID	Module	Test Scenario	Expected Result
TC19	Admin	Add a new medicine	Medicine appears in the catalogue of the user app
TC20	Admin	Open an admin screen using a normal user account	Access is denied

Table 7.3.1 Test Cases of PharmaCare

7.4 VALIDATION

Validation ensures that only correct and meaningful data is accepted by the application. The following validation checks are implemented at the client side in Flutter and are repeated at the server side by Firebase Security Rules.

- **Form Validation:** e-mail format, minimum password length, mobile number length and mandatory fields are checked before registration, login and address submission.
- **Search and Cart Validation:** empty search text is ignored, quantity cannot be less than one, and a medicine that is out of stock cannot be added to the cart.
- **Prescription Validation:** only supported image formats within the allowed size are accepted, and a medicine that requires a prescription cannot be ordered without a verified prescription.
- **Payment Validation:** an order is confirmed only after the payment gateway returns a successful status; failed or cancelled payments leave the order unconfirmed.

- **Reminder Validation:** the reminder time, days and dates are checked so that the end date is not earlier than the start date.
- **Location Validation:** the module checks that the location permission is granted and that GPS is enabled before searching for pharmacies.

7.5 QUALITY ASSURANCE

Quality assurance activities were carried out to make sure that the application is reliable, secure and easy to use. The quality attributes that were considered are described below.

- **Code Quality:** the code is organised module-wise, follows Dart naming conventions and is checked using the Flutter analyzer to remove warnings and unused code.
- **Performance:** images are compressed before upload, lists load data in small batches and Firestore queries are restricted to the required fields to keep the application responsive.
- **Security:** authentication is handled by Firebase Authentication, Firestore and Storage Security Rules restrict data access to the owner and the administrator, and sensitive keys are not written in the application code.
- **Usability:** the interface follows Material Design guidelines with clear labels, simple navigation and readable text so that it can be used by people of all age groups.
- **Reliability:** reminders are scheduled locally and rescheduled after a device restart so that they do not depend on the network, and stock updates use transactions to avoid wrong quantities.
- **Compatibility:** the application is checked on different Android versions and screen sizes to confirm that the layout and features work correctly.

7.6 DEFECT TRACKING AND RESOLUTION

Defects found during testing were recorded with the module name, the steps to reproduce the problem and the expected and actual results. Each defect was corrected by the developer responsible for the module, and the related test cases were executed again to confirm the correction. Regression testing was also performed on the neighbouring modules, so that a correction in one module did not affect the working of another. This cycle of testing, correction and retesting was repeated until the planned test cases produced the expected results.

7.7 SUMMARY OF TESTING

The testing, validation and quality assurance activities confirm that the modules of PharmaCare work as intended and that the application handles invalid inputs, permission denial and failed payments in a controlled manner. These activities increase the confidence that the application is dependable for managing medicines and prescriptions.

CHAPTER 8
CONCLUSION AND FUTURE SCOPE

8.1 CONCLUSION

PharmaCare – Smart Medicine Care Mobile Application provides a simple, effective and reliable solution for purchasing medicines and managing daily medication routines. By integrating online medicine ordering, prescription upload and verification, order tracking, medicine reminders and nearby pharmacy services, the application offers a complete digital healthcare solution in a single platform.

The use of Flutter and Firebase makes the application easy to develop, scalable and suitable for academic as well as real-world use. The project achieves its objective of improving medication adherence and the accessibility of pharmacy services, and it demonstrates how mobile technology can make everyday healthcare more convenient for patients, caregivers and pharmacists.

8.2 FUTURE SCOPE

PharmaCare has the potential to be expanded with several advanced features to further improve the user experience and healthcare support. The following enhancements can be added in the future.

- An AI chatbot to answer medicine-related queries.
- Medicine expiry date alerts for the medicines stored by the user.
- Daily health tips to promote healthy habits.
- QR code-based medicine verification to check the authenticity of medicines.
- Tamil and English language support for wider accessibility.
- Online doctor consultation within the application.
- Emergency medicine ordering with priority delivery.
- Order history with invoice download.

- Dark mode support for comfortable use at night.
- Voice search for medicines, which will help elderly and visually impaired users.

With continuous development, the application can evolve into a complete digital health companion that offers personalised and intelligent medicine management for users of all age groups.

APPENDICES

APPENDIX A - Source code

```
package com.pharmacare.app;

import android.content.Intent;
import android.graphics.Bitmap;
import android.net.Uri;
import android.os.Bundle;
import android.text.Editable;
import android.text.TextWatcher;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.EditText;
import android.widget.ImageView;
import android.widget.Spinner;
import android.widget.TextView;

import androidx.activity.result.ActivityResultLauncher;
import androidx.activity.result.contract.ActivityResultContracts;

import java.io.File;
import java.util.ArrayList;
import java.util.List;
import java.util.Locale;

/** Place an order for a medicine, optionally with a live photo. */
public class OrderActivity extends BaseActivity {

    private DatabaseHelper db;
    private List<DatabaseHelper.Medicine> meds = new ArrayList<>();
    private Spinner spMedicine;
    private EditText etQty, etAddress;
    private TextView tvTotal, tvPhotoHint;
    private ImageView ivPhoto;
    private File photoFile;
    private String photoPath = "";
    private long uid;

    private final ActivityResultLauncher<Uri> camera =
            registerForActivityResult(new ActivityResultContracts.TakePicture(), ok -> {
                if (ok && photoFile != null) {
                    photoPath = photoFile.getAbsolutePath();
                    Bitmap bm = PhotoHelper.load(photoPath, 600);
                    if (bm != null) {
                        ivPhoto.setImageBitmap(bm);
                        ivPhoto.setVisibility(View.VISIBLE);
                        tvPhotoHint.setVisibility(View.GONE);
                    }
                }
            });

    @Override
    protected void onCreate(Bundle b) {
        super.onCreate(b);
        if (!requireLogin()) return;
        setContentView(R.layout.activity_order);
        setupBack("Place Order");
        db = new DatabaseHelper(this);
        uid = Session.userId(this);

        spMedicine = findViewById(R.id.spMedicine);
        etQty = findViewById(R.id.etQty);
        etAddress = findViewById(R.id.etAddress);
        tvTotal = findViewById(R.id.tvTotal);
        tvPhotoHint = findViewById(R.id.tvPhotoHint);
        ivPhoto = findViewById(R.id.ivPhoto);

        meds = db.inStockMedicines();
        List<String> names = new ArrayList<>();
        for (DatabaseHelper.Medicine m : meds) names.add(m.name);
        spMedicine.setAdapter(new ArrayAdapter<>(this, android.R.layout.simple_spinner_dropdown_item, names));

        long pre = getIntent().getLongExtra("id", -1);
        for (int i = 0; i < meds.size(); i++) if (meds.get(i).id == pre) spMedicine.setSelection(i);

        DatabaseHelper.User u = db.getUser(uid);
        if (u != null && u.address != null) etAddress.setText(u.address);

        spMedicine.setOnItemSelectedListener(new AdapterView.OnItemSelectedListener() {
```

```
        @Override public void onItemSelected(AdapterView<?> p, View v, int pos, long id) { updateTotal(); }
        @Override public void onNothingSelected(AdapterView<?> p) { }
      });
      etQty.addTextChangedListener(new TextWatcher() {
        @Override public void beforeTextChanged(CharSequence s, int a, int c, int d) { }
        @Override public void onTextChanged(CharSequence s, int a, int c, int d) { }
        @Override public void afterTextChanged(Editable s) { updateTotal(); }
      });

      findViewById(R.id.btnPhoto).setOnClickListener(v -> {
        try {
          photoFile = PhotoHelper.newPhotoFile(this, "orders");
          camera.launch(PhotoHelper.uriFor(this, photoFile));
        } catch (Exception e) {
          toast("Could not open the camera");
        }
      });
      findViewById(R.id.btnPlace).setOnClickListener(v -> place());
      updateTotal();
    }

    private int qty() {
      try { return Integer.parseInt(etQty.getText().toString().trim()); } catch (Exception e) { return 0; }
    }

    private DatabaseHelper.Medicine selected() {
      int pos = spMedicine.getSelectedItemPosition();
      return (pos >= 0 && pos < meds.size()) ? meds.get(pos) : null;
    }

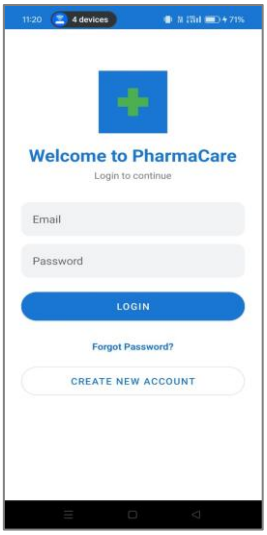
    private void updateTotal() {
      DatabaseHelper.Medicine m = selected();
      double total = m == null ? 0 : m.price * Math.max(qty(), 0);
      tvTotal.setText(String.format(Locale.getDefault(), "Total: ₹%.2f", total));
    }

    private void place() {
      DatabaseHelper.Medicine m = selected();
      int q = qty();
      String addr = etAddress.getText().toString().trim();
      if (m == null) { toast("No medicine available"); return; }
      if (q < 1) { etQty.setError("Enter quantity (1 or more)"); etQty.requestFocus(); return; }
      if (addr.isEmpty()) { etAddress.setError("Enter delivery address"); etAddress.requestFocus(); return; }

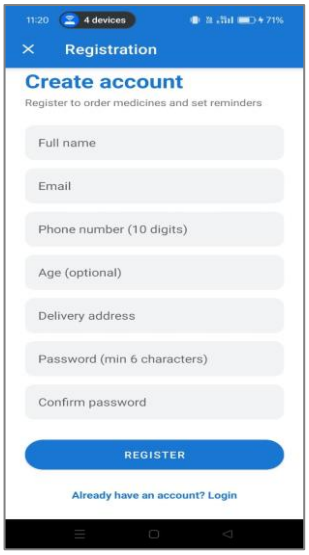
      String photo = photoPath.isEmpty() ? m.photo : photoPath;
      db.addOrder(uid, m.id, m.name, q, m.price * q, addr, photo);
      toast("Order placed successfully!");
      startActivity(new Intent(this, OrderHistoryActivity.class));
      finish();
    }
}
```

APPENDIX B

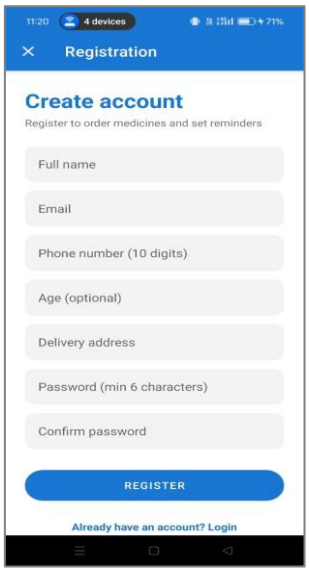
SCREENSHOTS



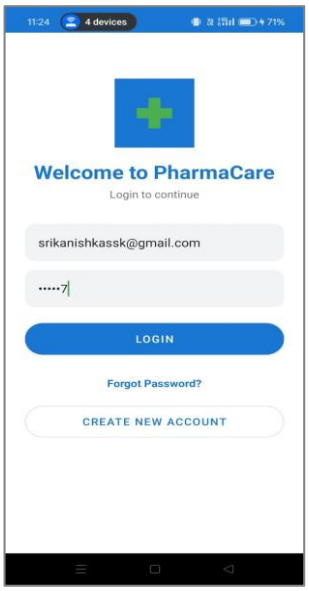
B.1 Login Page



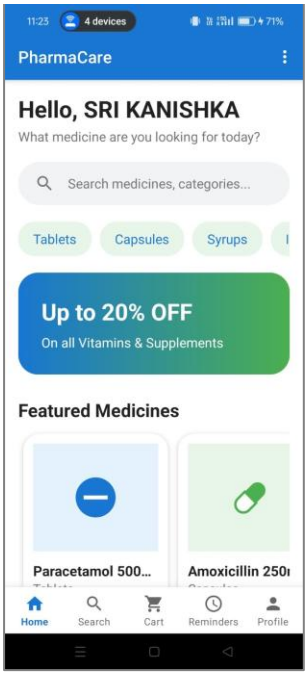
B.2 Registration Page



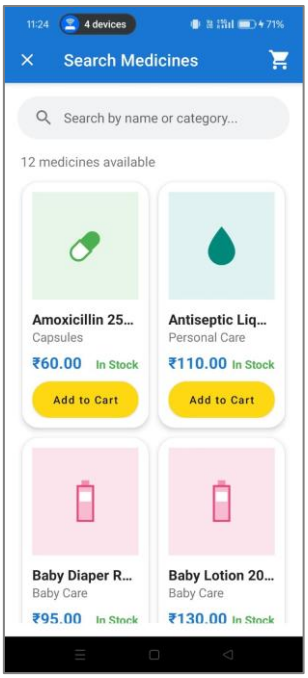
B.3 Registration Page (Scrolled View)



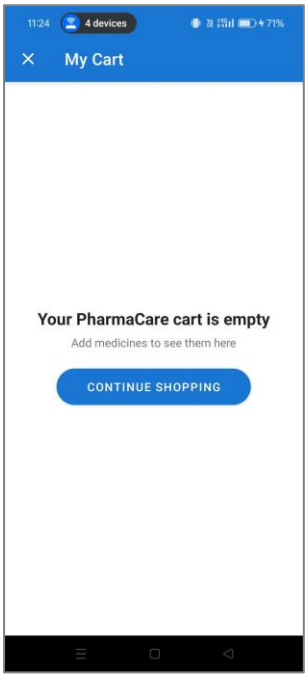
B.4 Login Page with Credentials Entered



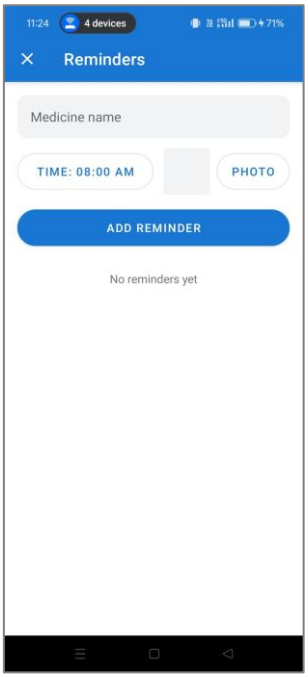
B.5 Home Page



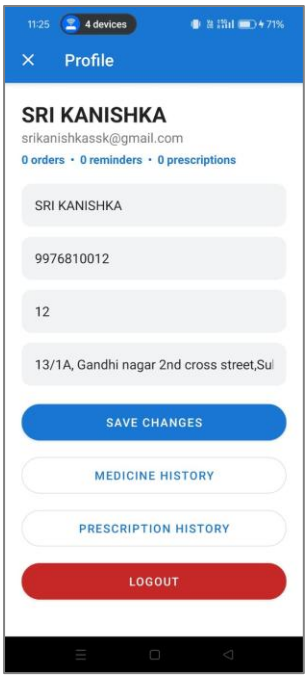
B.6 Search Medicines Page



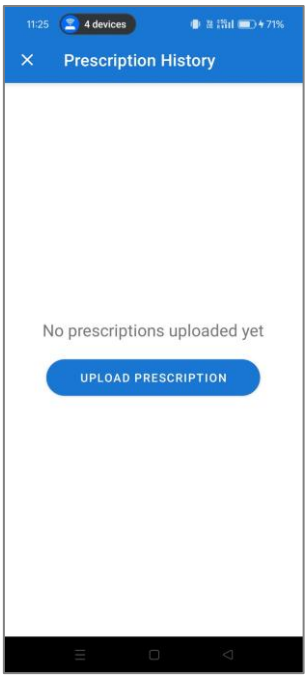
B.7 My Cart Page



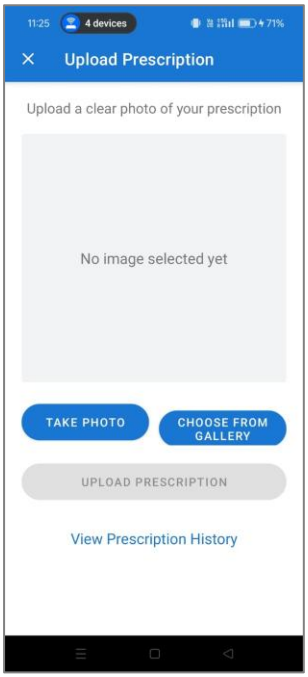
B.8 Medicine Reminders Page



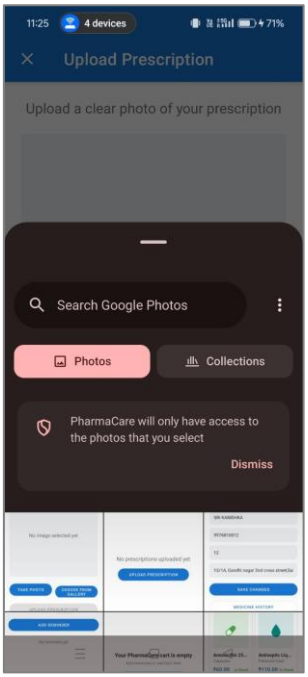
B.9 User Profile Page



B.10 Prescription History Page



B.11 Upload Prescription Page



B.12 Selecting a Prescription Image from the Gallery

REFERENCES

[1] Android Developers Documentation. Notifications, alarms and runtime permissions in Android. Available: https://developer.android.com

[2] L. Dayer, S. Heldenbrand, P. Anderson, P. O. Gubbins and B. C. Martin, “Smartphone medication adherence apps: Potential benefits to patients and providers,” Journal of the American Pharmacists Association, vol. 53, no. 2, pp. 172–181, 2013.

[3] Firebase Documentation, Google. Firebase Authentication, Cloud Firestore, Cloud Storage and Cloud Messaging. Available: https://firebase.google.com/docs

[4] Flutter Documentation, Google. Building cross-platform applications with Flutter and Dart. Available: https://docs.flutter.dev

[5] Google Maps Platform Documentation. Maps SDK, Places API and Directions. Available: https://developers.google.com/maps

[6] M. N. Kamel Boulos, S. Wheeler, C. Tavares and R. Jones, “How smartphones are changing the face of mobile and participatory healthcare: An overview, with example from eCAALYX,” BioMedical Engineering OnLine, vol. 10, art. 24, 2011.

[7] L. Osterberg and T. Blaschke, “Adherence to medication,” New England Journal of Medicine, vol. 353, no. 5, pp. 487–497, 2005.

[8] E. Sabaté, Ed., Adherence to Long-Term Therapies: Evidence for Action. Geneva, Switzerland: World Health Organization, 2003.

[9] K. Santo, S. S. Richtering, J. Chalmers, A. Thiagalingam, C. K. Chow and J. Redfern, “Mobile phone apps to improve medication adherence: A systematic stratified review,” JMIR mHealth and uHealth, vol. 4, no. 3, e132, 2016.

[10] K. Stawarz, A. L. Cox and A. Blandford, “Beyond self-tracking and reminders: Designing smartphone apps that support habit formation,” in Proc. 33rd Annual ACM Conference on Human Factors in Computing Systems (CHI ’15), Seoul, South Korea, 2015, pp. 2653–2662.